

Salman Haider

GIS Professional | Spatial Analyst | ArcGIS & Python Automation

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SUMMARY

GIS Professional and Spatial Analyst with MGIS training at the University of Calgary and applied GIS experience across Canada, Australia, and Pakistan. Skilled in ArcGIS Pro, spatial analysis, cartography, remote sensing, Python/ArcPy automation, web GIS, and spatial databases. Delivered 105+ franchise territory mapping projects and built practical GIS tools ranging from natural language ArcGIS Pro geoprocessing assistants to GTFS transit-access analysis and PostGIS routing applications. Strong focus on turning spatial data into clear maps, repeatable workflows, and decision-support outputs.

CORE TECHNICAL SKILLS

GIS Platforms, Web GIS & Spatial Databases: ArcGIS Pro, ArcGIS Online, ArcGIS Enterprise/Portal, ArcGIS Server, ArcGIS Field Maps, ArcGIS Dashboards, ArcGIS StoryMaps, ArcGIS Experience Builder, QGIS, Global Mapper, PostGIS, PostgreSQL, MS SQL, GeoServer, Leaflet, Mapbox GL JS, Mapbox Studio, MapTiler, Tippecanoe, Docker, Mango Maps.

Programming, Automation: Python, ArcPy, SQL, JavaScript, HTML/CSS, Flask, GeoPandas, GDAL, FME, Google APIs, Git/GitHub, workflow scripting.

Spatial Analysis, Cartography, Remote Sensing & Data Analysis: Network analysis, OD cost matrix, suitability modeling, MCDA, raster/vector analysis, spatial QA/QC, geocoding, demographic analysis, territory design, map layouts, Google Earth Engine, ENVI, ERDAS Imagine, image classification, spatial statistics, data visualization, Microsoft Office.

WORK EXPERIENCE

Franchise Ready

GIS Coordinator (Contract)

Nov. 2021 – Aug 2024

Sydney, Australia (Remote)

- **Delivered 105+ franchise territory mapping projects** using ArcGIS Pro, demographic datasets, competitor/POI data, and client-specific territory design criteria.
- Designed balanced franchise territories using variables such as population, income, competitor presence, target demographics, travel distance, and service areas to support market planning and site-selection decisions.
- Automated repetitive GIS preparation, web scraping/geocoding, and spatial processing tasks with **Python and ArcPy**, improving consistency and reducing manual analysis time.
- Prepared client-facing map products, territory summaries, and branded web map deliverables using ArcGIS Pro and **Mango Maps**.
- Maintained spatial data quality, projection consistency, and attribute accuracy so expansion recommendations were based on reliable and current datasets.

Greenage Services Pvt. Ltd

GIS Associate

Apr. 2021 – Nov. 2021

Islamabad, Pakistan

- Created cadastral and land-record maps using **ArcMap** for a government land record digitization project with strict accuracy requirements.
- Processed high-resolution satellite imagery, prepared GIS databases, and supported 3D visualization outputs for planning and project communication.
- Applied **QA/QC checks** to spatial layers, attributes, and map layouts to support reliable project deliverables.

EDUCATION

University of Calgary

Master of Geographic Information Systems (MGIS)

Aug. 2024 – May. 2026

Calgary, AB, Canada

- Graduate GIS training focused on spatial analysis, urban GIS, cartography and geovisualization, spatial databases, remote sensing, web mapping, network analysis, and GIS automation.
- **Final MGIS project: Geo Copilot** - natural-language geoprocessing assistant for ArcGIS Pro using LLMs, ArcPy, active map context, and reviewable execution workflows

Pir Meher Ali Shah Arid Agriculture University

B.Sc Geoinformatics

Sep. 2016 – Oct. 2020

Rawalpindi, Pakistan

- Developed a GIS and remote sensing-based solution as part of final year project for identifying **rainwater harvesting sites** in the Potohar region, enhancing sustainable agriculture practices.

SELECTED PROJECTS

Geo Copilot: Natural Language GIS Tool for ArcGIS Pro

ArcGIS Pro | ArcPy | Python | LLMs | MGIS Capstone

- Developed a custom ArcGIS Pro script tool (“Geo Copilot”) that enables users to perform geoprocessing tasks using natural language queries, integrating **Large Language Models (GPT-5 and Llama 3.1)** with ArcPy workflows.
- Designed structured reasoning frameworks to translate user intent into executable spatial workflows.
- Evaluated model performance across 36 geospatial test scenarios, achieving up to **91.7%** function selection accuracy and **83.3%** parameter accuracy with GPT-based workflows, and analyzed failure cases related to projection conflicts and multi-step reasoning.

Transit-to-Table: Calgary Fresh Food Access

ArcGIS Pro | GTFS | Network Analyst | OD Cost Matrix | Urban GIS

- Built a **GTFS-based multimodal transit network** using Calgary Transit schedule data, street centerlines, ArcGIS Public Transit Data Model, and walking + transit travel-time costs.
- Modeled DA-level access from Dissemination Area origin points to fresh food destinations assembled from OSM grocery/supermarket points and Calgary food hubs using a 45-minute weekday OD cost matrix.
- Found **12 dissemination areas** with no access and **32 dissemination areas** with poor access. These areas represented **33,212 residents**, showing that Calgary has transit-based fresh-food access but that access quality is uneven

Logistics Routing MVP

PostGIS | pgRouting | Flask | Docker | Leaflet | OpenStreetMap

- Built a self-hosted web GIS routing prototype using OSM road data, PostGIS/pgRouting, Flask API, Docker containers, and Leaflet.
- Implemented start/end point selection, shortest-time vs shortest-distance routing, blocked polygon zones, traffic-delay points, route alternatives, and GeoJSON route display.
- Prepared SQL workflow for road network creation, topology, routing costs, one-way handling, and spatial indexing for route calculation.